



AI.Data Lab Project Kickoff

Request

If you are participating but have not been added to the Canvas site for the program, please see one of us!

- We'll add you right now!

Introductions

- AI.DataLab Faculty/External Leads
 - Dr. Kevin McAlister
 - Dr. Tyler Cook
 - Dr. Huy Huynh
 - Dr. Allison Wasson
 - Nicole Torno
- Student Leads
 - Dalton Lowery
 - Miriam Brem
 - Kelsey Olesen
 - Caroline Zeipel
 - Paulo Condori Pinedo

Program Overview & Expectations

Introductory Discussion Board Post (5%)

- The program has grown!
 - More people = Good
 - More people = Harder to know everyone
- This program has a lot of interest
 - From Emory admin (particularly the Provost) and partners who are looking for interns/potential employees etc.
- Before September 17th, create a post on the appropriate discussion board that has:
 - Photo
 - Name
 - Major and Interesting Facts
- Will use this info to create a directory of participants that we can then pass on to interested groups!

Program Overview & Expectations

- Teams of 3
 - Your assigned team will be your team for the semester!
 - A mixture of backgrounds, some variation in skills, but trying to create cohesive teams
 - Diversity is a good thing! New insights come from those who see things in a different way.
 - Expectations: patience, participation, and respect
 - Fail to do even one of these and we will boot you from the program
 - Everyone has something to learn. Everyone struggles. That's how we get better!

Program Overview & Expectations

- Split into two subprograms this year
 - Sandbox – intended for students who are beginners (we all were once) and/or students who are looking for a setting where they can learn and grow as a data scientist with more start-up time and lower deliverable pressures
 - Foundry – intended for more experienced students who will come into the project with good programming skills, strong quant backgrounds, and some previous experience working on data science projects
- The split is largely unimportant
 - Intended to show our commitment to place more experienced students on teams
 - Also a way for you to show growth on your resumes when you talk about this program!
 - One or two projects will be **Foundry only** due to the necessary background skills and investment needed to provide infrastructural backend tools

Program Overview & Expectations

- Project Structure
 - Each project will have multiple teams working on the same general question(s)
 - Broad enough that there will be different approaches, question interpretations, subquestions, etc. for each team
 - Organizing team will be frequently checking in to see team progress
 - Attempting to ensure no teams end up in the same place
 - Project mentors will check in with teams frequently and provide guidance and suggestions if we think that the projects are too similar
 - Be creative! Try weird things. Fail frequently.
 - The best ideas come after a string of failures. Goal isn't **just** to have a successful project – it's to have a **great** project idea and do the best you can with what you've got.

Program Deliverables

- Academic poster
- Open-source code base

Results will be presented at an end of semester showcase on **December 10th, 6-8:30!**

Each team will have the opportunity to present their work directly to project sponsors, interested academic community from Emory, and external folks.

Project #1: Identifying Low Income Housing Need

- Housing has become an increasingly large portion of the average Atlantan's spending
- Rent hikes have outpaced incomes
- Funding for affordable housing is difficult to obtain.
- Shortage of affordable housing has displaced residents
- Housing shortages affect everyone



Project #1: Identifying Low Income Housing Need



What Is LIHTC?

Federal program offering 10-year tax credits to incentivize private investment in affordable rental housing

Finances ~90% of all new affordable housing development in the U.S



Eligibility Rules

20/50 Rule: 20% of units for tenants earning 50% or less of the area median income

40/60 Rule: 40% of units for tenants earning 60% or less of the area median income

Must meet long-term affordability and construction standards



How It Works

State agencies (e.g., Georgia DCA) allocate credits competitively

Developers claim credits over 10 years in exchange for setting aside rent-restricted units for 30 years



Program Goals

Increase the supply of affordable rental housing

Promote economic diversity in housing markets

Catalyze private-sector investment

Project #1: Identifying Low Income Housing Need

Scoring Category	Description	Max Points
Extended Affordability Commitment	Points for committing to affordability beyond the minimum 30-year term.	6
Minority- and Women-Owned Business Engagement	Engagement of MWBE firms in development and construction.	2
Favorable Financing	Use of below-market financing to support project feasibility.	5
Compliance Performance	Good standing in past DCA projects; no significant noncompliance.	10
Integrated Supportive Housing	Set-asides for persons with disabilities in integrated settings.	5
Readiness to Proceed	Demonstration of site control, zoning, utility service letters, and readiness to close on financing.	13
Deeper Targeting/Rent/Income Restrictions	Rents targeted significantly below 60% AMI, or income tiering	3
HCV and PHA Notices	Demonstrated coordination with Housing Choice Voucher and Public Housing Authority programs.	2
Desirable/Undesirable Activities	Proximity to assets (schools, healthcare) and distance from undesirable features (landfills)	20
Community Transportation Options	Accessibility to public transit and walkability.	6
Quality Education Areas	Located within zones of high-performing schools.	3
Revitalization/Redevelopment Plans	Participation in or proximity to recognized plans (e.g., Choice Neighborhoods).	10
Stable Communities	Located in areas with stable homeownership and school attendance patterns.	10
Community Designations	Located within an area with a specific state or federal designation	10
Phased Development	Part of a multi-phase development (physically adjacent phases, shared amenities).	4
Previous Projects	Successful past experience in Georgia tax credit projects.	5
Housing Needs Characteristics	Project in a high-growth area in terms of population or employment.	10
Economic Development Proximity	Close to major job centers, identified economic zones.	1
Mixed Income Developments	Incorporates market-rate units.	1
Historic Preservation	Rehabilitation of a certified historic structure.	2
Enriched Property Services	On-site services that support residents' health, education, or economic mobility.	2
DCA Community Initiatives	Supports DCA-identified strategic community priorities.	2
Preservation Scoring Criteria	For properties with expiring subsidies or in critical condition, promoting long-term affordability.	6

Project #1: Identifying Low Income Housing Need

<https://kmcalist682336.github.io/sitelineLIHTC/>

Project #1: Identifying Low Income Housing Need

- We have a way to produce the score for any point in the Metro 11 counties
 - Some quality problems, but 90ish percent accurate is our best guess
- Now, let's use it to answer questions and propose some action!
- New addition: measure **affordable housing need**
 - Not as straightforward as you may think
 - Lots of renters, lower income/rent ratio
 - But, can we **move** renters to newer high QOL areas?
 - Add a commute. But, better schools. Better job opportunities. Better grocery access.
 - Consult existing literature and other non-profits to come up with a way to measure!

Project #1: Identifying Low Income Housing Need

- Project Goal: Use the LIHTC scoring tool to tell/recommend something to the affordable housing non-profit community
 - Pretty open ended. We're providing the data tool for one half of the problem. You and your team come up with the other.
- Rules:
 - Open source only
 - Actionable recommendations/insights
 - Be creative – a lot of work has already been done. Do something different!

Project #1: Identifying Low Income Housing Need

- Potential Deliverables:
 - A web app that shows locations in the metro 11 where LIHTC score is high and affordable housing developments would be plausible and useful
 - Identifying city and county owned properties that aren't being used that have high LIHTC scores that could be used for development
 - A creative algorithm for identifying housing need
 - A data-driven policy case for developing in Alpharetta vs. Atlanta or vice versa
- The options are endless. We're just looking for high quality creativity.
 - The same thing has been used for a long time to mixed effect. Why not propose something different!

Project #2: AI Policy in K-12 Schools

- School districts across the country are grappling with designing AI policies for students without knowing what actually works.
 - At least 30 states have recently published K-12 AI guidance with very little existing evidence about the effects of such guidance on student learning.
- This project requires students to map what K-12 AI policy currently looks like at the federal, state, and district level, and then explore questions related to these findings.
- Project sponsor: local school district that is very interested in what you might discover!

Project #2: AI Policy in K-12 Schools

- What do metro-Atlanta district AI policies actually say, and how do they compare to other districts? What are the gaps, challenges, or other shortcomings within such policies?
- Do AI policies differ systematically between well-resourced and under-resourced districts, and if so, does that widen existing educational gaps?
- Is there credible evidence that AI policies change student behavior or improve educational outcomes? What are the best ways we could gather such evidence, given how new this all is?

Project #2: AI Policy in K-12 Schools

- Nearly all of it is free and public!
- **Policy trackers:** AI for Education state tracker, TeachAI Toolkit, Education Commission of the States, EdSAFE
- **District documents:** BoardDocs and Simbli board policy, student handbooks, acceptable use policies, GA Open Records requests
- **State and legislative:** GaDOE “Leveraging AI in the K-12 Setting,” NCSLAI database, Georgia General Assembly
- **Survey and panel data:** RAND American School District Panel, CDT annual surveys, EdWeek Research Center, Pew
- **Administrative and outcome:** GOSA data downloads, GaDOE Insights, GA•AWARDS, NCES and the Civil Rights Data Collection

Project #2: AI Policy in K-12 Schools

- **Searchable policy database:** federal, state, and district AI policies in one filterable dashboard
- **Comparative matrix:** how AI policies in metro-Atlanta districts differ from each other and policies in other districts
- **Evidence gap map:** showing which AI policy claims have research behind them, and which have none (or have research suggesting they're ineffective!)
- **Policy recommendations:** what should be changed and why, including evidence to support each proposed change
- **Future directions:** implications or suggestions for future research, limitations and questions your project leaves open, etc.

Project #3: Sports Analytics

- The Modern Sports Analytics Lab is a joint effort between Emory's Data and Decision Sciences department and CAIL
- Goal: Provide a place where students can learn about **real** sports analytics and work on cutting edge projects related to football (the American variety), baseball, and cricket.
- Modern sports analytics is more ML than sports expertise
 - Understanding ways to handle the complex data that is produced

Project #3: Sports Analytics

- Students will work in teams of 3 on a question of their own creation related to one of three sports (that we can provide some knowledge and expertise about):
 - Football (College and/or Professional) (Me)
 - Baseball (Dalton)
 - Cricket (Riyan – when he's able to join since he's 12 hours different...)
- Data is endless (except for biometrics and real-time positioning data that are largely proprietary due to expense in capturing)
 - Use existing data to answer some question or provide some new knowledge about something in one of these sports that you understand and love

Project #3: Sports Analytics

- Football:
 - Changes in 1st round draft success before/after modern NIL policies
 - Fantasy football projections
 - Quantifying and understanding route styles (aggregate and individual) by receivers in the NFL
 - NFL teams **really** don't know how to leverage modern Gen AI tools in modern sports analytics!

Project #3: Sports Analytics

- Baseball:
 - At bat outcome predictions (strike/walk/whiff)
 - Optimal infield/outfield alignment given offensive scenario
 - Projecting minor leaguer performance in first MLB year
 - Optimal lineup solvers
 - Biometrics work using open source data sets

Project #3: Sports Analytics

- Cricket:
 - Park/Lineup effects
 - Optimal batter ordering
 - Game-scenario bowler changes
 - Ball tracking/Bowl Shaping using Youtube videos of matches (consistent bowler wicket coordinate rotations to measure post-ground movement)
 - Identifying lower league talent early

Project #3: Sports Analytics

- Deliverables:
 - A Github page that shows off your work (easy to do with AI coding assistants now!!!)
 - Open source codebase (need not be fully public)
- The goal here is to provide you with other people interested in the sport to work with and our expertise to help you do a good sports analytics project!

Project #4: Businesses in Atlanta with InvestATL

- Two subprojects this year!
 - Both related to business health in Atlanta

Small Business Health

Small Business Health

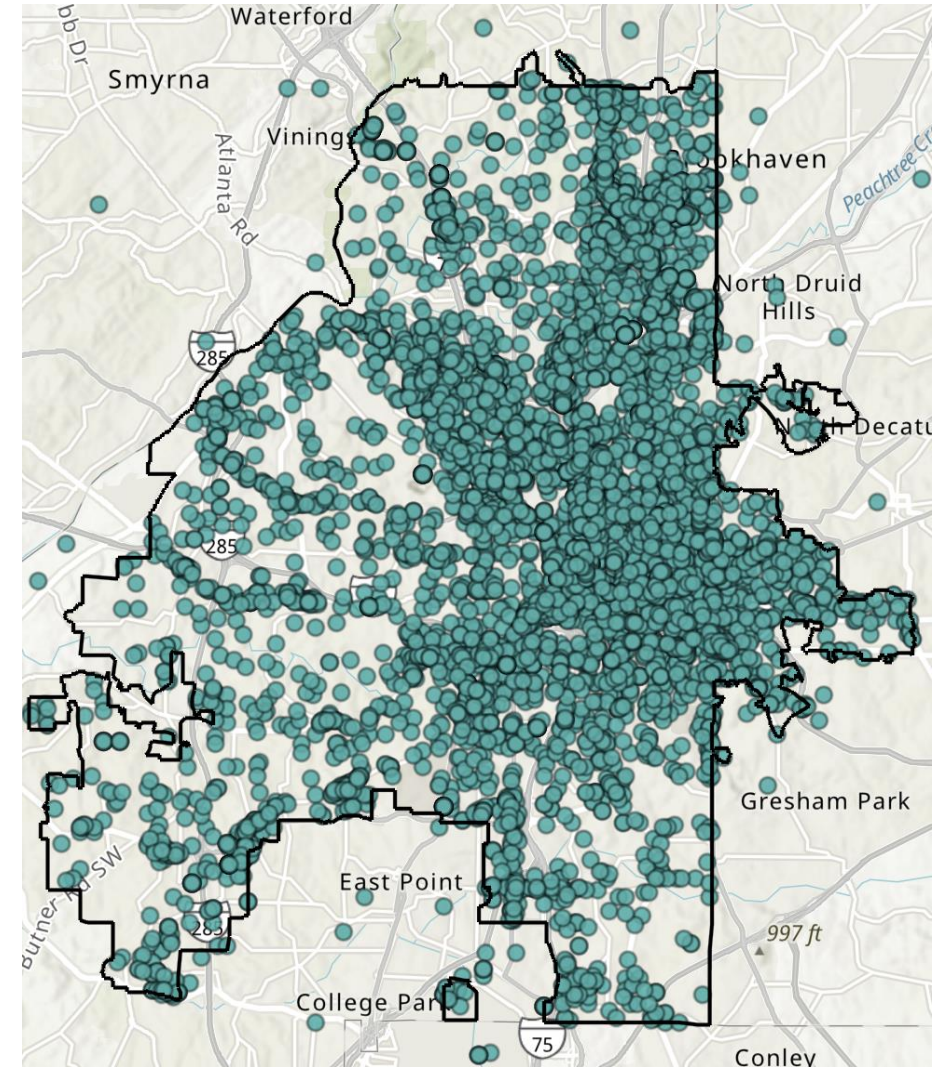
- Invest Atlanta has a broad range of small business support tools, but understanding what the community needs takes more than surveys and meetings
- How do we know where small businesses are thriving, what kind of support is needed, and how the community is impacted by these businesses?



2025 recipients Strikeout Wings and Molly Sanyour Ceramics

Small Business Health

- Key results:
 - Analysis of where small businesses are(n't) growing in different industries
 - Analysis of commercial space availability vs need and opportunities for support
- Startup data:
 - Registered businesses in Atlanta
 - Tax parcel records (including building type, vacancy, costs)
 - Publicly-owned parcels
 - Previous small business funding and program scale



All registered businesses in Atlanta, 2026

Business Competitiveness

Atlanta's Competitiveness

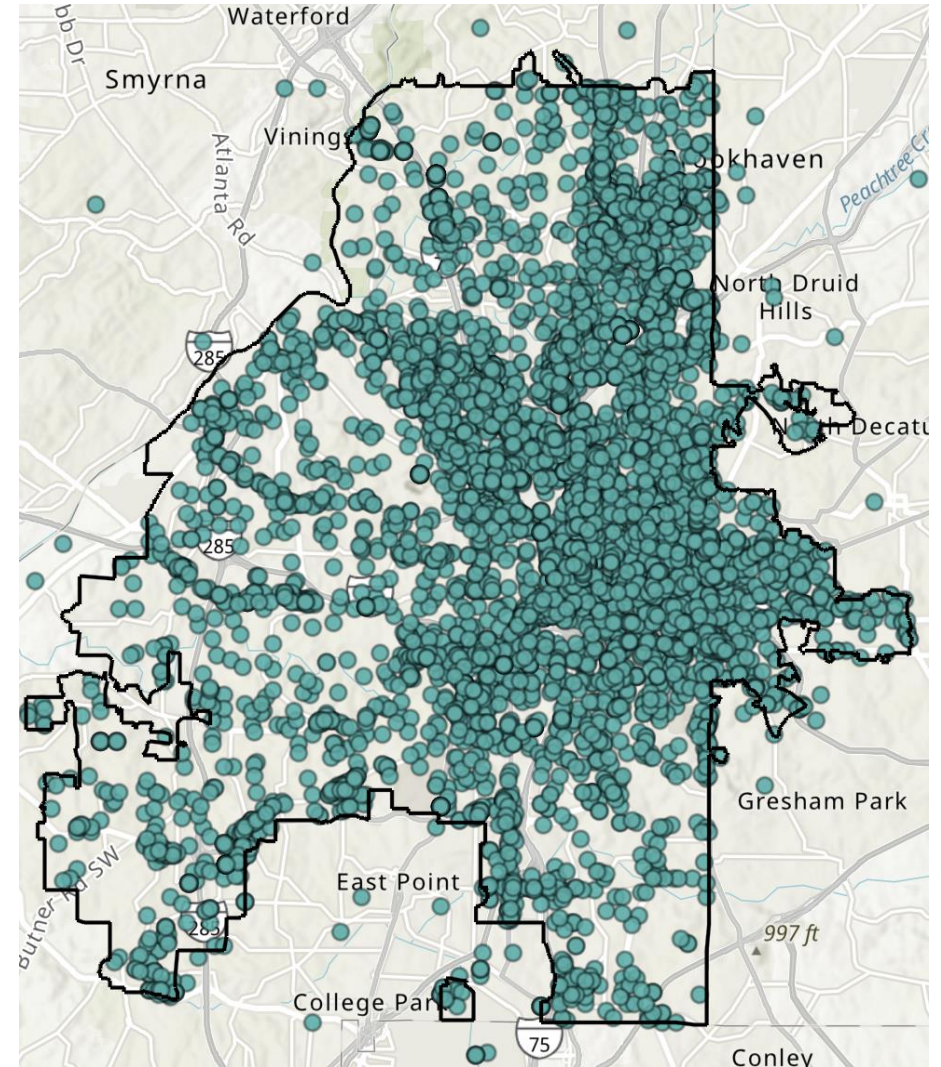
- Invest Atlanta plays a key role in major business relocations and expansions as an economic development agency
- Atlanta is home to more than 15 Fortune 500 companies, and these headquarters locations are essential to our economy
- In order to stay at the top of our region, our incentives, commercial real estate, workforce development, and amenities all have to shine to major corporations



2025 Fortune 500 deals from Invest Atlanta

Business Competitiveness

- Key results:
 - Where is Atlanta lagging in business attraction and retention?
 - Which incentives in other regions should be implemented or expanded here?
- Startup data:
 - Registered businesses in Atlanta
 - Invest Atlanta business incentives
 - Tax parcel records (including building type, vacancy, costs)
 - BLS data regarding industry trends and business growth/loss in Atlanta and other regions



All registered businesses in Atlanta, 2026

Project #5: An AI Advisor for Emory STEM programs

- Quick note: we will have, at most, 4 teams working on this project!
- Trying to navigate Emory's intricate STEM major environment can be quite tough:
 - What majors should I consider given my career goals?
 - What majors lead to the widest variety of career outcomes since I don't know what I want to do with my life?
 - I know that I want to do something related to data science. What majors should I consider and what classes can I take in my first year that won't limit my options later?

Project #5: An AI Advisor for Emory STEM programs

- Claude and other general AI tools are decent at this
- More specialized tools exist but they are largely outdated
- Advances in **multi-agent systems** and a better understanding of when dense embeddings vs. Pure nearest neighbor lookups should be used means that there's a lot of room for improvement!
- A tool that would be wildly popular with students and advisors.

Project #5: An AI Advisor for Emory STEM programs

- Goal: Develop an AI agent that can be used to assist students and advisors
 - **Curated data layer:** Correctness is more important than completeness
 - **Time specific logic:** Ability to answer abstract questions about plans to complete a major given time limitations
 - **Reasonable abstraction based on data:** Ability to reasonably answer questions that aren't directly covered in the data (e.g. What majors should I consider if I want to go into quant finance? Based on placement data, which departments have alumni networks in the NYC consulting scene? How can I choose a major to go along with my BBA major that will increase my chances of getting into a top MFE program after graduation?)
 - **Ability to integrate answers to common questions:** Easy process to ingest question/answer pairs that can be provided by advisors that address common questions that clearly are not well answered by existing documentation.
 - And a lot of other potential stuff.

Project #5: An AI Advisor for Emory STEM programs

- Using whatever truthful data you can collect, build something that addresses one or more of these things
 - I don't think that there's enough time to build, test, iterate, etc. And do all of those things well
 - Figure out your niche, solve it, and create a tool that does that part well
 - All teams should create a solid ground truth data layer and use vector DB matching tech to ensure that answers are truthful! Incorrect answers are immediate disqualifiers.
 - Clear testing framework and answer quality batteries

Project #5: An AI Advisor for Emory STEM programs

- Tech frameworks
 - Start by developing a local tool that leverages fully open weight models on the backend
 - FAISS DBs for ground truth vector storage
 - Open source embedding models
 - Proof of concept using local resources
 - Once you have a reasonable proof of concept, transfer system to use Center OpenAI resources via OpenAI API and develop a user friendly front-end

Project #5: An AI Advisor for Emory STEM programs

- Deliverables
 - Poster describing process and tech solutions
 - Working demo
 - Short tech doc that describes the approach and the tech stack as well as any developer decisions made throughout the process
 - A working Github (private is fine as long as you ultimately give the Center access to the materials at the end of the semester – we'll keep it private!)
- Post-semester
 - Combine the ideas into one working tool and (hopefully) create an open release for campus usage with Emory credential gating
 - All that make meaningful contributions will have creator billing at the end

Project Overview

- 5 different projects (with subprojects within) that cover drastically different topics
 - Different substance
 - Different tech tools
 - Different resume skills
- All have been designed to be good resume builders!
 - Can be buzzworded up to help you in your career goals
 - Variety of different tech skills even within the projects

Project Overview

- AI Assistant project will only be open to Foundry teams
- All other projects are open to all!
 - The idea is that there can be a variety of projects at different skill levels
 - All good, just different approaches
- Key distinction: being placed on a Foundry team means that you can assume that everyone on your team is ready to start the technical part of the project from day 1
 - Higher technical expectations, higher deliverable expectations
 - Slightly higher expected project commitment – accomplishing more means committing a little more time!

Project Overview

- I expect that roughly 1/3 of teams will be Foundry
 - All others classified as Sandbox
- Create a path to show growth on your resume!
- Within Sandbox teams, we'll try to create teams that have similar skill levels

Survey

- Plenty of time for questions
- Before leaving the room, fill out the preference survey linked on the following slide
 - Basic demographic info
 - How do you see yourself contributing to a project team?
 - Short skills/experience assessment
 - Project preference ordering
- For the skill questions, please answer these on your own without the use of AI tools or other assistance
 - Using this to judge previous experience as best as we can
 - Want to set everyone up for success
 - Be honest. If you don't know, then select the don't know option.
 - There's a place for all skill levels in this program. Knowing it will allow us to create the best possible situation for you!

Preference Survey



[https://tinyurl.com/AIDLF26
PreferenceSurvey](https://tinyurl.com/AIDLF26PreferenceSurvey)